

NITHIN ESWARAPPA

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INTRODUCTION

I have researched and developed solutions for projects involving multimodal vision sensors, optimized for embedded platforms, performing data fusion, object detection, and segmentation. My recent research interests include image synthesis, image translation,, multimodal / multisource image registration and fusion, vision foundation models and 3D perception using vision only sensors. Total Experience: 7+ years.

EDUCATION

Visvesvaraya Technological University

Master Of Technology In Intelligent Systems

December 2015

Bengaluru, India

Visvesvaraya Technological University

Bachelor of Engineering in Electronics and Communication

June 2008

Bengaluru,
India

PUBLICATIONS

2025 "BiFIS: Bi-modal Fused Image Synthesis Using Dense Distillation with Multilayer Feature

Alignment on a Vision Foundation Model", Eswarappa N, Jeevan KM, Waldekar S, Vivek BK,
Submission *Planned* George K.

2025 "TriFIS: Trimodal Fused Image Synthesis framework and A Visible-Infrared-SAR Benchmark
Dataset", Eswarappa N, Jeevan KM, Waldekar S, Vivek BK, George K.

*Planned
Submission*

2025 **IEEE TENCON**, "HyLapDFN: A Hybrid Approach for Infrared-Visible Image Fusion using

Laplacian Pyramid and Decoder Fusion Network," Eswarappa N, Jeevan KM, Waldekar S, Vivek [preprint](#)
BK, George K. *Oral*

2025 **IEEE SPACE**, *Conventional vs Deep-Learning Methods for Visible and Infrared Image Fusion: A* [Oral](#)
Comparison. Eswarappa N, Jeevan KM, Waldekar S, Vivek BK, George K.

2025 **IEEE SPACE**, Trimodal Image Fusion Framework for a Visible-Infrared-SAR Benchmark Dataset. [Oral](#)
Eswarappa N, Jeevan KM, Waldekar S, Vivek BK, George K.

2024 **IEEE CONECCT**, *A Colour-Focused Visible and Infrared Image Fusion Framework for Aiding*
[Oral](#) *Human Perception*. Eswarappa N, Waldekar S, Jeevan KM, Vivek BK, George K.

RESEARCH EXPERIENCE

RESEARCH ASSOCIATE, Multi Sensor Image Fusion Lab (DRDO grant), GITAM University

- Research on trimodal image fusion using Visible, Infrared, and SAR images for aiding

Oct 2023 - Now

human operator and multimodal image fusion for object detection and classification. Bengaluru, India

GRADUATE STUDENT, Centre for Intelligent Systems, PESIT

- Master's Thesis: "*Appearance based terrain learning and classification for Oct 2013 - Apr 2015 autonomous mobile robot navigation*", advised by Prof Prabhakar Mishra. *Bangalore, India*

PROFESSIONAL MEMBERSHIP

Member, IEEE (Institute of Electrical & Electronics Engineers)

Since 2024

AWARDS

GATE Fellowship, Electronics and Communication Engineering,
Indian Institute of Technology, Madras, Graduate Aptitude Test in Engineering,
Ministry of Human Resource Development (MHRD), Government of India (GOI).

March 2011

MENTORING EXPERIENCE

BOct 2023 - Nowengaluru, India

Multi Sensor Image Fusion Lab (DRDO grant), GITAM University

- Supervising and mentoring a junior researcher on multimodal image fusion using conventional and deep-learning based methods.
- Supervising and mentoring 10 students (1 Master's and 9 undergraduate students) for fulfilment of engineering degree projects.
- Master's Thesis – Multi-modal Image registration for robust image fusion.
- Undergraduate Projects: Disparity estimation using stereo image pairs, Deep learning based visible and thermal infrared image fusion, Optimization of image fusion methods for real-time applications on embedded platform, Nvidia Jetson platform.

PROFESSIONAL EXPERIENCE

Senior Firmware Engineer - Robotics, Vimaan.ai

Apr 2020 - Jul 2022,

Bengaluru, India

- Developed a method for absolute localization with 6-DOF, in indoor environments, using camera pose estimation with the use of ArUco markers.
- Developed a method for relative pose estimation using stereo data from Intel RealSense depth cameras for drone localization in indoor environments.
- Automated recharging of drone batteries using a battery management system for continuous autonomous drone operations.

Apr 2015 – Jun 2018,

Member of Technical Staff, Tonbo Imaging

Bengaluru, India

- Conducted research and development of methods, to correct motion blur in narrow FOV images, from dual electro Optonics (thermal IR and visible) on a continuously rotating (360°/sec) multi sensor imaging platform.
- Developed methods for stitching 360° panoramic image from line of sight stabilized images for long range passive RADAR, used for surveillance and automatic threat detection from air and sea.

- Designed and developed a synchronization (camera sensor – Vsync) mechanism, and optical motor control protocol for motion blur correction on an embedded platform.
- Designed and developed Target-pointing (Geo-pointing) capability for Pan Tilt Unit (PTU) mounted imaging systems on embedded platforms, integrating multiple subsystems such as GPS, digital magnetic compass, and ground based RADAR.
- Developed algorithms for rolling shutter correction, image blur correction, non uniformity correction, and automatic gain correction for thermal images.
- Automated a tool for Optical centre estimation without calibration and field of view estimation for multiple zoom steps.
- Developed command control interface for multi-sensor active range gated camera and arranged and conducted meetings with the vendors for technology - knowledge transfer and module integration into systems.
- Developed a multi-interface (Serial/UDP/TCP) multithreaded communication architecture on embedded Linux for remote command and control of multi sensor surveillance platforms (Camera TIR, Visible, PTU, GPU, Compass, LRF). *Jul 2008 - Apr 2009,*

Systems Engineer, Siemens Information Systems Ltd.

Bengaluru, India

Developed functionalities and addressed change requests in the service module of Syngo – Siemens Health care, Medical Imaging software.

TECHNICAL SKILLS

- **Programming:** C/C++, Python, Matlab, Bash, Git, LaTeX
- **Lib / Middleware:** OpenCV, Robot Operating System (ROS)
- **Other Tools:** QGIS, SNAP Toolbox, QT, Visual Code
- **Embedded systems:** Nvidia Jetson Platforms, Arduino, BeagleBone, DM365, DM8168, IMX6
- **Operating System:** Ubuntu, Windows